Dice Simulator Assignment

https://gemini.google.com/app/81eec4938357e1b5

User prompt: what does the teacher want for this assignment?

Response: Based on the image provided, your teacher wants you to program a **"Dice Simulator" (骰子模擬器)** program (likely using Java Swing given the reference to `JLabel`).

Here is the translation and breakdown of the required specifications:

Topic (題目)

Implement a **"Dice Simulator" (骰子模擬器)**.

Specifications (規格)

1. **Window Settings (視窗設定)**
   - **Title:** Set to 骰子模擬器.
   - **Size:** `400×320` pixels.
   - **Behavior on close:** Terminate/exit the program when closed (`EXIT_ON_CLOSE`).
   - **Startup position:** Center the window on the screen when opened (`setLocationRelativeTo(null)`).
2. **Center Label (中央 JLabel)**
   - Include a `JLabel` in the center of the window to display the current rolled dice number.
   - Font size: **60pt**.
3. **Bottom Button (下方按鈕)**
   - Add a button at the bottom labeled **擲骰子** ("Roll Dice").
4. **Rolling Logic (隨機產生與更新)**
   - Every time the button is clicked, randomly generate an integer between **1 and 6**.
   - Update the center label to display the newly rolled number.
5. **Top Statistics Label (視窗上方顯示)**
   - Display a status string at the top of the window with the format:

     `已擲 N 次，總和 M，平均 X.XX`

     (e.g., "Rolled N times, total sum M, average X.XX", where average is formatted to 2 decimal places).
6. **Text Color Logic (點數顏色化)**
   - If the rolled value is **6**, change the text color to **Green**.
   - If the rolled value is **1**, change the text color to **Red**.
   - For all other values (**2, 3, 4, 5**), set the text color to **Black**.

User prompt: give me the code and the file name is Exercise2.java

Response: Here is the complete Java Swing code for your assignment, structured with the requested class and file name `Exercise2.java`:

```
import javax.swing.*;
import java.awt.*;
import java.awt.event.ActionEvent;
import java.awt.event.ActionListener;
import java.util.Random;

public class Exercise2 extends JFrame {
```

```
    private JLabel topLabel;
    private JLabel centerLabel;
    private JButton rollButton;

    private int rollCount = 0;
    private int totalSum = 0;
    private Random random = new Random();

    public Exercise2() {
        // 1. Window Settings
        setTitle("骰子模擬器");
        setSize(400, 320);
        setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
        setLocationRelativeTo(null); // Center on screen
        setLayout(new BorderLayout());

        // 5. Top Label (Statistics)
        topLabel = new JLabel("已擲 0 次，總和 0，平均 0.00", SwingConstants.CENTER);
        topLabel.setFont(new Font("SansSerif", Font.PLAIN, 14));
        add(topLabel, BorderLayout.NORTH);

        // 2. Center Label (Dice Result)
        centerLabel = new JLabel("-", SwingConstants.CENTER);
        centerLabel.setFont(new Font("SansSerif", Font.BOLD, 60));
        add(centerLabel, BorderLayout.CENTER);

        // 3. Bottom Button
        rollButton = new JButton("擲骰子");
        rollButton.setFont(new Font("SansSerif", Font.PLAIN, 18));
        add(rollButton, BorderLayout.SOUTH);

        // 4 & 6. Button Event Handling (Rolling Logic & Color Logic)
        rollButton.addActionListener(new ActionListener() {
            @Override
            public void actionPerformed(ActionEvent e) {
                // Generate random number between 1 and 6
                int diceValue = random.nextInt(6) + 1;

                // Update counters
                rollCount++;
                totalSum += diceValue;
                double average = (double) totalSum / rollCount;

                // Update top statistics text
                topLabel.setText(String.format("已擲 %d 次，總和 %d，平均 %.2f", rollCount, totalSum, average));

                // Update center label text
                centerLabel.setText(String.valueOf(diceValue));

                // Set text color according to the rules
                if (diceValue == 6) {
                    centerLabel.setForeground(Color.GREEN);
                } else if (diceValue == 1) {
                    centerLabel.setForeground(Color.RED);
                } else {
                    centerLabel.setForeground(Color.BLACK);
                }
            }
        });
    }

    public static void main(String[] args) {
        // Run GUI construction on Event Dispatch Thread
        SwingUtilities.invokeLater(() -> {
            new Exercise2().setVisible(true);
        });
    }
}
```

Would you like help setting up the compilation environment or adding any extra features to the interface?